

PART II. COMPONENT RESPONSIBILITY SPECIFICATION

15 points

Each row is a component shown in Figure 1. Responsibilities are deliberately non-overlapping: recognition, correctness, sequencing, counting, and persistence are five separate jobs held by five separate components.

Component	Responsibility	Input	Processing / Function	Output	Depends On
Main Menu Controller	Starts a session and captures the player's nickname	Nickname text, Play button	Validates the entry and stores it, then loads the level map	Saved nickname, scene change	Nickname Manager
Level Select Controller	Presents the four environments as a journey map	Taps on map nodes	Shows each node locked or unlocked and blocks locked ones	Scene load for the chosen level	Level Progress
Scene Objective Controller	Drives the ordered spoken-word objectives of one scene	Objective list (word, variants, prompt, audio, effect)	Shows the current prompt, sets the target word, advances on a correct answer, raises the per-word effect	Prompt display, correct-word events, scene-complete event	Voice Command
Voice Command	Bridges the game to the device speech recognizer	Microphone audio, current target word	Starts and restarts recognition, receives recognized text, asks Pronunciation Checker to judge it	OnAnswerChecked (text, wasCorrect)	Device speech recognizer
Pronunciation Checker	Decides whether the spoken text counts as the target word	Recognized text, target word, accepted variants	Normalises both strings, checks exact and variant matches, then a length-scaled edit distance	Correct / not correct	None (pure C# utility)
Player Movement	Moves and poses the character	Calls from objective effects	Applies movement and swaps the character's sprite set	Updated position and appearance	Unity physics
Pick Up Item	Transfers a scene prop onto the character	A correct answer for a pick-up word	Disables the prop and swaps to the matching character art set	Updated scene and character art	Player Movement
Scene Fader / Room Transition	Moves the player between scenes without a hard cut	Scene-complete or exit trigger	Fades to black, loads the next scene by name, fades back in	Next scene loaded	Unity SceneManager
Run Score Counter	Counts this scene's voice attempts into the level total	OnAnswerChecked events	Adds one attempt, and one correct answer when applicable, to Run Score	Updated running total	Voice Command, Run Score
Run Score	Holds one playthrough's totals across scene loads	Attempt results	Keeps correct and attempt counts as static values; reset at level start and after a save	Correct count, attempt count	None (static class)
Session Score Tracker	Turns a finished level into a saved record and shows it back	Scene-complete event, Run Score totals, nickname	Builds the record, requests the save, then requests recent records and writes them into the completion panel	Saved record, on-screen score and recent results	Supabase Client, Run Score, Nickname Manager
Supabase Client	Performs the actual database calls	Record to insert, row limit to fetch	POSTs one row and GETs recent rows over HTTPS, attaching the anon key headers	Success / error, list of records	Supabase Config, Supabase REST API
Supabase Config	Supplies connection credentials without committing them	Project URL and anon key entered locally	Stores them in a git-ignored asset read at runtime	Endpoint and key	Unity Resources
Level Progress	Remembers which environments are unlocked	Level completion	Raises the highest unlocked level and persists it	Unlock state for the map	PlayerPrefs

Component	Responsibility	Input	Processing / Function	Output	Depends On
Nickname Manager	Remembers who is playing	Nickname from the main menu	Caches and persists the nickname, with a friendly default	Nickname for display and for the saved record	PlayerPrefs